

EXPERIMENT 39

Install Hadoop 2.x and Configure the NameNode and DataNode

AIM

To install Hadoop 2.x on a Linux system and configure the NameNode and DataNode for a pseudo-distributed, single-node Hadoop cluster.

REQUIREMENTS

- Ubuntu Linux system with hduser configured (see Experiment 38)
- Java Development Kit installed and JAVA_HOME set
- Hadoop 2.x binary distribution (tar.gz) downloaded from the Apache archive
- Terminal access with sudo privileges

PROCEDURE

1. Download the Hadoop 2.x tarball from the official Apache Hadoop download mirror.
2. Extract the tarball and move the extracted folder to /usr/local/hadoop.
3. Change ownership of the /usr/local/hadoop directory to hduser and the hadoop group.
4. Edit ~/.bashrc to set HADOOP_HOME, JAVA_HOME and update the PATH variable.
5. Edit hadoop-env.sh to set the JAVA_HOME environment variable for Hadoop.
6. Configure core-site.xml with the default filesystem (fs.defaultFS) pointing to the NameNode.
7. Configure hdfs-site.xml with the replication factor and the NameNode and DataNode storagedirectories.
8. Configure mapred-site.xml and yarn-site.xml for the MapReduce and YARN framework.
9. Format the HDFS NameNode using the hdfs namenode -format command.

IMPLEMENTATION

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hduser@localhost: /usr/local/hadoop

hduser@localhost:~$ wget https://downloads.apache.org/hadoop/common/hadoop-2.10.2/hadoop-2.10.2.tar.gz
Saving to: 'hadoop-2.10.2.tar.gz'
100%[=====] 366M saved

hduser@localhost:~$ tar -xvzf hadoop-2.10.2.tar.gz -C /usr/local/
extracting hadoop-2.10.2/ ...

hduser@localhost:~$ sudo mv /usr/local/hadoop-2.10.2 /usr/local/hadoop

hduser@localhost:~$ nano ~/.bashrc
export HADOOP_HOME=/usr/local/hadoop
export PATH=$PATH:$HADOOP_HOME/bin:$HADOOP_HOME/sbin
export JAVA_HOME=/usr/lib/jvm/java-8-openjdk-amd64

hduser@localhost:~$ nano $HADOOP_HOME/etc/hadoop/core-site.xml
<property><name>fs.defaultFS</name>
<value>hdfs://localhost:9000</value></property>

hduser@localhost:~$ nano $HADOOP_HOME/etc/hadoop/hdfs-site.xml
<property><name>dfs.replication</name><value>1</value></property>
<property><name>dfs.namenode.name.dir</name>
<value>/usr/local/hadoop/hdfs/namenode</value></property>
<property><name>dfs.datanode.data.dir</name>
<value>/usr/local/hadoop/hdfs/datanode</value></property>

hduser@localhost:~$ hdfs namenode -format
Formatting using clusterid: CID-6f1e2a3b-....
Storage directory /usr/local/hadoop/hdfs/namenode has been successfully formatted.
```

Figure 1: Illustrative terminal view — Hadoop installation and NameNode/DataNode configuration.

RESULT

Hadoop 2.x was installed successfully, the NameNode and DataNode were configured in hdfs-site.xml, and the HDFS NameNode was formatted without errors.